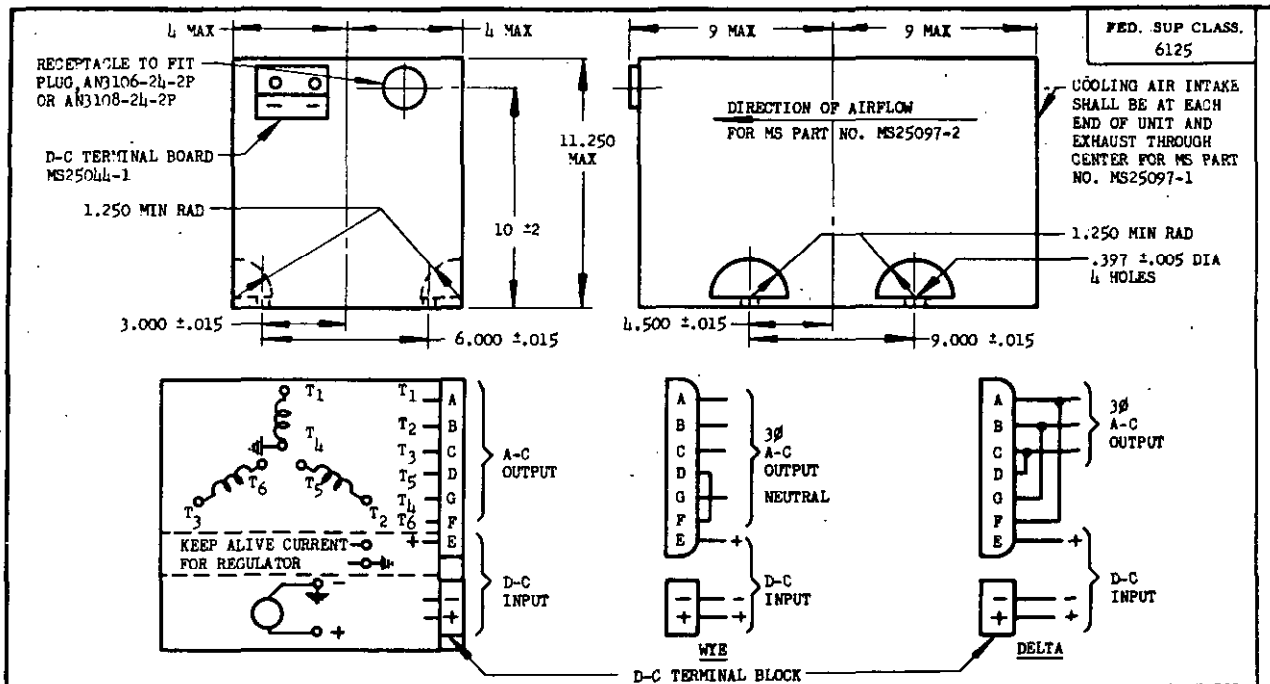
NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

FED. SUP CLASS. 6125



PHASE ROTATION A, B, C

CONNECTIONS EXTERNAL TO THE INVERTER NECESSARY FOR WYE (115/200 VOLT) AND DELTA (115 VOLT) CONNECTIONS FOR 1Ø CONNECTION, SEE (a)

MS PART NO.	COOLING AIRFLOW
MS25097-1	DUPLEX
MS25097-2	SIMPLEX

	3Ø	1Ø
INPUT: VOLTS, DC	28	28
OUTPUT: VOLTS, AC	115/200	115
FREQUENCY, CPS	400	400
VA TO 50,000 FEET ALTITUDE	2500	2500
VA TO 65,000 FEET ALTITUDE	1250	1250
PF MIN LAG, PERCENT	90	90
PF MIN LEAD, PERCENT	95	95
MIN OUTPUT VOLTATE AT 200 PERCENT RATED CURRENT	50/85	50
MIN EFFICIENCY: PERCENT AT RATED LOAD, SEA LEVEL, AND 25° ±15°C	50	45
WEIGHT, POUNDS, MAX	62	62

FOR ANY COMBINATION OF CONDITIONS LISTED BELOW, THE OUTPUT VOLTAGE AND FREQUENCY SHALL REMAIN WITHIN THE INDICATED LIMITS.

CONDITIONS	LIMITS	
	VOLTAGE	FREQUENCY, CPS
INPUT VOLTAGE VARIED BETWEEN 26 AND 29 VOLTS LOAD VARIED FROM 0 TO FULL LOAD AMBIENT TEMPERATURE-ALTITUDE VARIED WITHIN THE RANGE ESTABLISHED BY CURVE I OF MS33513, EXCEPT THAT THE AMBIENT TEMPERATURE FOR THIS UNIT SHALL NEVER EXCEED 85°C. THE UNIT SHALL DELIVER RATED OUTPUT TO 50,000 FEET AND 50 PERCENT MIN RATED POWER TO 65,000 FEET. LOAD POWER FACTOR VARIED FROM .90 LAG TO .95 LEAD.	LOW 110/192 HIGH 120/208	LOW 390 HIGH 410

(a) THIS INVERTER DESIGN SHALL BE SUCH THAT IT MAY BE LOADED EITHER 3Ø OR 1Ø BUT NOT SIMULTANEOUSLY. WHEN USED AS A SINGLE-PHASE INVERTER, THE UNIT SHALL BE CONNECTED IN DELTA AND THE LOAD CONNECTED TO THE TERMINALS A(T_1) AND B(T_2). WHEN USED 3Ø, THE LOADS PER PHASE SHALL NOT DIFFER BY MORE THAN 10 PERCENT OF THE INVERTER RATING.

DIMENSIONS IN INCHES.
THE AIRCRAFT MANUFACTURER SHALL ALLOW CLEARANCE FOR THE MAXIMUM DIMENSIONS SHOWN ON THIS DRAWING FOR INSTALLATION.

THE WEIGHT AND DIMENSIONS SHOWN ON THIS DRAWING SHALL INCLUDE THE WEIGHT AND DIMENSIONS OF ALL AUXILIARY APPARATUS NECESSARY TO MAKE THE UNIT CONFORM TO THE PROCUREMENT SPECIFICATION.
THE MS DESIGNATION SHALL NOT BE SHOWN ON INVERTERS UNTIL APPROVAL HAS BEEN RECEIVED FROM THE QUALIFYING ACTIVITY AND THE INVERTER FULLY MEETS THE REQUIREMENTS OF THE SPECIFICATION AND DRAWING.
A MEANS OF ADJUSTING THE VOLTAGE SHALL BE ON THE CONNECTOR END AND ABOVE THE HORIZONTAL CENTERLINE OF THE INVERTER.
PERMISSION OF THE PROCURING ACTIVITY SHALL BE OBTAINED FOR APPLICATION OF THIS UNIT OUTSIDE OF THE SPECIFIED LIMITS.

TWO PLACARDED PIN JACKS SHALL BE PROVIDED ON THE CONNECTOR END AND ABOVE THE HORIZONTAL CENTERLINE OF THE UNIT BETWEEN WHICH THE VOLTAGE BETWEEN A(T_1) AND G(T_4) CAN BE MEASURED. THESE PIN JACKS SHALL ACCEPT A PROD PIN .078 ±.002 INCHES IN DIAMETER.

FOR NAVY APPLICATION, INVERTERS WITH EITHER SIMPLEX OR DUPLEX COOLING AIRFLOW CONFIGURATION ARE ELECTRICALLY AND MECHANICALLY INTERCHANGEABLE WITH EACH OTHER.

THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

(D) **CANCELED** AFTER 10 FEBRUARY 1965. USE MS17404. INTERCHANGEABILITY RELATIONSHIP: MS17404 PARTS CANNOT ALWAYS REPLACE THE CANCELED PARTS IDENTIFIED BY THE SAME DASH NUMBERS. THE CANCELED MS25097 PARTS CANNOT REPLACE THE MS17404 PARTS.

REVISED (A) 20 Jan 56 (B) 15 Nov 56 (C) 12 Dec 58 (D) 10 Feb 65
APPROVED 27 Oct 54

CUSTODIANS	OTHER INT.
Navy - WP USAF - 82 (D)	A - N - AF -

MILITARY STANDARD

INVERTER, 2,500 VA, 3/1 PHASE, CLASS B

MS25097 (ASG)

PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 1 OF 1
MIL-I-7032		

DD FORM 672-1
1 OCT 52

Source: https://assist.dla.mil — Downloaded: 2026-10-23T15:03Z
Check the source to verify that this is the current version before use.